SPOORTHY KUMBASHI RAGHAVENDRA

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EDUCATION

Texas A&M University | College Station, TX, USA **Aug 2025 – May 2027**
Master of Computer Science – CGPA: ***4.0***

PES University | Bengaluru, India **Aug 2019 – May 2023**
B.Tech in Electronics and Communication Engineering – ***Gold Medalist*** *– CGPA:* ***3.9/4.0***

TECHNICAL SKILLS

Languages : Python, **Java**, **SQL**, Ruby, MATLAB, HTML, CSS
AI/ML & Data : PyTorch, **TensorFlow**, Keras, Scikit-learn, **Hugging Face Transformers**, OpenCV, NumPy, Pandas, **LangChain**
Frameworks & Tools : Spring Boot, Django, Flask, **React**, Ruby on Rails, **Docker**, **Kubernetes**, Jenkins, Git

WORK EXPERIENCE

Jefferies LLC – Technology Summer Associate | Jersey City, NJ, USA **Jun 2026 – Aug 2026**

- Developed an **agentic system** to diagnose trade settlement failures by mapping dependency chains across system states and inventory data, with dynamic clustering by root cause and priority to maximize resolution with minimal manual action.
- Modeled a **Markov Decision Process (MDP) recommendation engine** constrained by business logic to predict optimal resolution paths, integrating a reinforcement learning loop to iteratively update the policy model based on operator overrides.

JP Morgan Chase & Co. – Software Engineer I | Bengaluru, India **Jun 2023 – Jul 2025**

- Designed and deployed scalable **Spring Boot microservices** for processing 250-300k daily volumes of Foreign Exchange trades worth $10-15B, reducing operational risk and maintenance costs while improving system resilience.
- Built reusable **Java libraries** to support multi-**MQ** consumers and **Kafka** Ack/Nack handling between services, enhancing observability across platforms and reducing message debugging time by 4-5 hours per service.
- Converted the **REST APIs** between the UI and backend to **GraphQL** reducing frontend data processing complexity and improving load time by ~100ms.

JP Morgan Chase & Co. – Software Engineering Intern | Bengaluru, India **Feb 2023 – May 2023**

- Developed a real-time trade flow monitoring platform with dynamic pod scaling and log streaming using **React** and **Spring Boot**.
- Improved visibility across 20+ microservices and cut root cause analysis and **resolution time by 40%**.

Bosch – Machine Learning Intern | Bengaluru, India **Jun 2022 – Jul 2022**

- Engineered an **NLP pipeline** with a custom position-based vectorizer to structure symbolic test reports, boosting equation parsing accuracy by ~30%.
- Trained **Random Forest/SVM** models on EV testing reports for automated test generation, cutting manual effort by ~50%.

PROJECTS

Constraint-Aware Financial Disclosure Search Engine

- Built a hybrid **BM25 + FinBERT search engine** over SEC 10-K filings, integrating **GLiNER**-based constraint extraction and OpenSearch to retrieve evidence passages from S&P 100 disclosures.
- Implemented **penalty-based re-ranking** across temporal, geographic and sector dimensions to improve retrieval precision.

Critic-Guided Agentic RAG for Question Answering

- Developed an agentic **RAG** pipeline with a **zero-shot LLM critic** that iteratively evaluates retrieved context and rewrites queries.
- Reduced hallucinations, **outperforming single-pass RAG** on multi-hop QA benchmarks like HotpotQA.

PUBLICATIONS

Deep Learning for Signal Detection in RSMA Receiver ***Lead Author*** *@ IEEE NCC 2023;* **DOI** ***10068068***

- Developed an **LSTM-based deep learning model** for wireless signal detection, using temporal sequence modeling to reduce bit error rates and outperform conventional signal detection algorithms in noisy, multi-user beyond-5G channels.

Deep Learning for Radiogenomic classification of Brain Tumor ***Lead Author*** *@ IEEE INDICON;* **DOI** ***10039760***

- Led research applying transfer learning with **ResNet** and **AlexNet** on 3D MRI scans to predict genetic biomarker status in brain tumor patients, achieving **82% accuracy** on the RSNA-MICCAI dataset via optimized slice extraction and augmentation.

HONORS AND AWARDS

- **Gold Medal –** Ranked first in department for overall academic performance.
- **Six Merit Scholarships –** Recognized with Prof. C.N.R. Rao and Dr. M.R.D. Scholarship for sustained academic excellence.